Haoran (Leo) Liu

MSc Robotics @ Georgia Tech
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EDUCATION

Georgia Institute of Technology — MSc in Robotics (College of Computing) (GPA 3.88/4.0) Aug 2025 – May 2027 (expected)

The Hong Kong University of Science and Technology — BSc in Mathematics, First Class Honours Sep 2021 – Jun 2025

PUBLICATIONS

[1] Z. Zhao, M. Leggiero, Y. Chen, **H. Liu**, Y. Wu, H. Xue, S. Zhan, Y. Zhao. VQActFlow: Vector-Quantized Action Mode Steering for Multi-Task Robot Manipulation. *IEEE Robotics and Automation Letters (RA-L), 2026.* arXiv

[2] **H. Liu***, et al. DLONet: Local-to-Global Consistent Occlusion Ordering for Multi-Cable Harness Perception. *Under review, IEEE RA-L*

[3] Y. Wu, L. Fan, Z. Li, ..., **H. Liu**, ..., (25 authors). RoboChrono: A Real Robot Benchmark for Streaming Task Understanding. *arXiv preprint, September 2026.* arXiv

*Equal contribution. Full author list will appear with the arXiv version.

RESEARCH EXPERIENCE

Research Intern, General Intelligence Machine (GIM) Jun 2026 – Present

- Core contributor to RoboChrono, a real-robot benchmark for streaming task understanding (39 scenarios, 34.7k instances, 18 VLMs evaluated): led VR-teleop data collection, the annotation/QA protocol, and the VLM evaluation harness.
- Operate GIM's dual-arm whole-workspace platform with VR teleoperation for data collection and policy evaluation.

Graduate Research Assistant, Georgia Institute of Technology (PI: Prof. Sehoon Ha) Jan 2026 – Present

- Built a real-to-sim-to-real pipeline (X-Sim style): phone video → COLMAP → 2D Gaussian Splatting → watertight mesh → SAPIEN/ManiSkill scene with CoACD collision and auto-generated URDF.
- Integrated Gaussian-splat environments into Isaac Lab and a Neuromeka NAMI platform (USD scene swap, Zenoh messaging) for sim-trained grasping policies on a mobile dual-arm robot.

Graduate Research Assistant, LIDAR Lab, Georgia Institute of Technology (PI: Prof. Ye Zhao) Aug 2025 – Present

- Co-author of VQActFlow (IEEE RA-L 2026): led LIBERO experiments, Unitree G1 teleop data and deployment, ALOHA-style bimanual experiments, and latency engineering.
- Reward modeling for VLA post-training: LoRA-finetuned Qwen3-VL-8B-Instruct for sub-task segmentation and success classification on LIBERO-Spatial; Qwen2.5-VL chain-of-thought reward labeling (criteria → description → analysis → rating).

PROJECTS

Vision & Autonomy Engineer, HKUST RoboMaster Team Sep 2022 – Oct 2023

- Built the auto-aim pipeline (OpenCV + YOLOv8 on industrial cameras) with predictive target tracking at 7–8 m, improving bullet efficiency by ~80%.
- Developed the autonomous Sentinel robot: ROS2 + FreeRTOS, LiDAR/IMU/camera sensor fusion for navigation and engagement.

OTHER RESEARCH EXPERIENCE

Shenzhen Shunhe Electric — Research Scientist Intern: Multi-view video temporal alignment with a Simone-style model and curriculum learning: 99% accuracy on 3-class and >90% on 9-class temporal-offset classification. Jun 2025 – Aug 2025

ETH Zurich — Research Assistant (remote): Positional-encoding scheme for lexicographic rule hierarchies in autonomous-vehicle decision making. Sep 2024 – Apr 2025

University of California, Berkeley — Research Assistant (remote): Graphlet-based bootstrapping for hypothesis testing against network null models. Apr 2024 – Sep 2024

SKILLS

ML: PyTorch, JAX, Hugging Face Transformers / PEFT (LoRA), GRPO / PPO, Flow matching & diffusion policies, LeRobot
Robotics: ROS 2, Isaac Lab, ManiSkill / SAPIEN, LIBERO, Unitree G1 / Z1, Kinova Gen3, Neuromeka NAMI, VR teleoperation
Perception / 3D: 2DGS / 3DGS, COLMAP, SAM 2, GTSAM, OpenCV, YOLO
Tools: Python, C++, Linux, Docker, Slurm (PACE), Blender, LaTeX